09.26

EXP NO : 11(a)- STATIC ROUTING

AIM :

To simulate and configure static routing with primary and backup routes using cisco packet tracer and verify connectivity between different networks.

PROCEDURE:

1. Open cisco packet tracer and create the given topology with router 0 and router 1, router 2, switches and PC's.
2. Configure the IP addresses on all the routers interfaces and PC's according to the given topology
3. Open the CLI of router 0 and enter configuration mode.
4. Configure static routes for networks 30.0.0.0/8 and 50.0.0.0/8 using routers as the primary route and router 2 as the backup route.
5. Configurate a separate host route for 30.0.0.1000/32 with router 2 as the primary route and router 1 as the backup route.
6. On router 1, configure static routes for networks 10.0.0.0/8 and 40.0.0.0/8 with primary and backup routes.
7. On router 2, configure static routes for networks 10.0.0.0/8 and 30.0.0.0/8.

8. Use the command : show ip route static to verify the configured static routes.

9. Use the tracer command from a PC to verify the path taken by packets to the destination network.

10. Disconnect the primary link between R0 and R and use traceart again to verify that the backup route becomes active.

11. Static routes can be removed using : no ip route.

12. Verify the routing table again using show ip route static.

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Router>
Router>enable
Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#
Router(config)#interface GigabitEthernet0/0/0
Router(config-if)# ip address 192.172.17.254 255.255.255.0
Router(config-if)# no shutdown
Router(config-if)# exit
Router(config)#
Router(config)#interface GigabitEthernet0/0/1
Router(config-if)# ip address 172.190.32.254 255.255.255.0
Router(config-if)# no shutdown
Router(config-if)# exit
Router(config)#
Router(config)#
Router(config)#
Router(config)#
Router(config)#
Router(config)#
Router(config)#
```

RESULT :

Thus, the routing with primary and backup routes was successfully configured and simulated using the cisco packet tracer.